

Introduction

Functions are one of the most fundamental concepts that occur everywhere in mathematics, statistics, and computer science. They provide the means to transport information from one set to another.

Learning Objectives

- Define a function using both the set-theoretic and traditional $f : A \rightarrow B$ notation, and in particular be able to talk about the data of a function and what functions represent.
- Define the image and preimage of a set under a function. Be able to compute and prove (pre-)images for relatively simple functions.
- Define injective, surjective, and bijective functions. Provide plain English explanations describing the characteristics of such functions.
- Be able to identify and prove whether a function is injective, surjective, or bijective.
- Prove simple results about injective, surjective, and bijective functions.

The Real Definition of a Function

Recall the following from our discussion of relations on sets.

Definition 1

If A and B are sets, a *binary relation* R between A and B is a subset $S_R \subseteq A \times B$. If $a \in A$ and $b \in B$, we say that a is *related* to b (written aRb) if $(a, b) \in S_R$. Moreover, we say that R is

- *left-total* if for every $a \in A$ there exists $b \in B$ such that aRb .
- *functional* if for every $a \in A, b_1, b_2 \in B$, if aRb_1 and aRb_2 then $b_1 = b_2$.

We didn't do much with these definitions at the time, but we will now. The following is the definition of a function. While it may look different than that with which you are familiar, it is nonetheless the same!

Definition 2

If A and B are two sets, a *function* f from A to B is a left-total, functional binary relation on A and B . In this case, A is said to be the *domain* of f , and B is the *codomain* of f .

Let's unravel what this means. If $A = \{2, 4, 6\}$ and $B = \{\heartsuit, \diamond\}$, then $S_f = \{(2, \heartsuit), (4, \heartsuit), (6, \diamond)\}$ defines a left-total and functional relation which we'll call f . But instead of writing $2f\heartsuit$, the notation with which you're more familiar is to write $f(2) = \heartsuit$. Similarly, $f(4) = \heartsuit$ and $f(6) = \diamond$. This allows us to interpret

what left-total and functional actually mean:

- left-totality says that a function must be defined for every element of its domain,
- functionality says that an element of the domain can only map to one element in the codomain (the “vertical line test”).

Functions are often written as $f : A \rightarrow B$, explicitly indicating the domain, the codomain, the name of the function, and the direction in which the function travels. However, **the codomain is not the range of f** : The codomain only indicates where the output of f lives. You’ve probably heard of the graph of a function, whose proper definition is

$$\Gamma(f) = \{(a, f(a)) : a \in A\} \subseteq A \times B.$$

Notice that this is exactly the set which defines the relation f !

Exercise 3 Let A and B be sets with $A \subseteq B$. Define a relation g on $A \times B$ by $S_g = A \times A \subseteq A \times B$. Does g define a function?

Even though this is the correct formal definition of a function, in practice mathematicians do not think about functions like this. Instead, we think of functions as machines that eat elements in A and produce elements in B . You may generally ignore the relational definition of a function, except when explicitly instructed otherwise.

Exercise 4 Suppose that X is an arbitrary set and $\emptyset$ is the empty set. Is there a function $f : \emptyset \rightarrow X$? Is there a function $g : X \rightarrow \emptyset$? Explain why or why not.

Exercise 5 Which, if any, of the following functions are equal to one another?

1. $f : \{0, 1\} \rightarrow \mathbb{R}, x \mapsto x$
2. $g : \{0, 1\} \rightarrow \mathbb{R}, x \mapsto x^2$
3. $h : \{0, 1\} \rightarrow \{0, 1\}, x \mapsto x$.

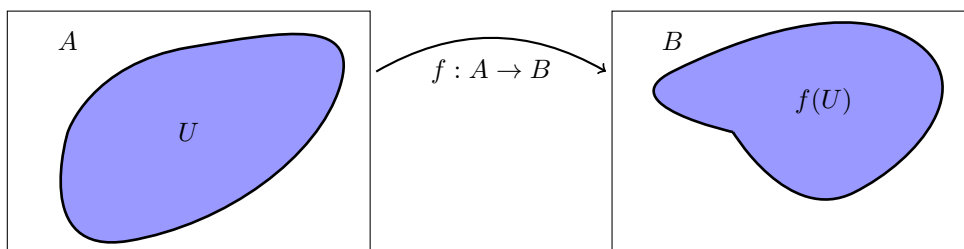


Figure 1: A function and its image set.

Images and Preimages

Definition 6

Let $f : A \rightarrow B$ be a function.

1. If $U \subseteq A$, then we define the *image* of U to be

$$f(U) = \{y \in B : \exists x \in U, f(x) = y\} = \{f(x) : x \in U\}.$$

2. If $V \subseteq B$, we define the *pre-image* of V to be

$$f^{-1}(V) = \{x \in A : f(x) \in V\}.$$

Note that despite being written as $f^{-1}(V)$, the preimage of a set does not say anything about the existence of an inverse function.

Example 7

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be specified by $f(x) = x^2$. Determine $f([-1, 2])$.

Solution. By definition, one has $f([-1, 2]) = \{f(x) : x \in [-1, 2]\}$, which we claim is $[0, 4]$. Indeed, let's show a double subset inclusion. Suppose that $y \in [0, 4]$ and let $x = \sqrt{y}$. We know that x exists since $y \geq 0$ and that x must live in $[0, 2] \subseteq [-1, 2]$. Thus $f(x) = y$, showing that $y \in f([-1, 2])$.

On the other hand, suppose that $x \in [-1, 2]$ so that $-1 \leq x \leq 2$. It follows that $0 \leq x^2 \leq 4$ (why?), so that $f([-1, 2]) \subseteq [0, 4]$. Both inclusions show that $f([-1, 2]) = [0, 4]$. ■

Example 8

Let $p : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be given by $p(x, y, z) = (x, y)$. If

$$D = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq 1\},$$

determine $p^{-1}(D)$.

Solution. While it's not a strictly correct picture, we can think of p as a projection map, taking a three-dimensional shape and squishing it into the xy -plane. If we think of p like this, then the pre-image of D is the set of things which get squished into D ; namely, the solid cylinder

$$C = \{(x, y, z) : x^2 + y^2 \leq 1\}.$$

Let's prove that $p^{-1}(D) = C$ via double subset inclusion.

($\supseteq$) Suppose that $c \in C$ and write $c = (x_0, y_0, z_0)$. By definition, we know that $x_0^2 + y_0^2 \leq 1$ and that z_0 can be anything. But then

$$p(c) = p(x_0, y_0, z_0) = (x_0, y_0) \quad \text{and} \quad x_0^2 + y_0^2 \leq 1,$$

so that $p(c) \in D$ by definition. Thus $C \subseteq p^{-1}(D)$.

($\subseteq$) Let $(x_0, y_0, z_0) \in p^{-1}(D)$, which means that $p(x_0, y_0, z_0) = (x_0, y_0) \in D$. Being an element of D means that (x_0, y_0) satisfies $x_0^2 + y_0^2 \leq 1$, and so $(x_0, y_0, z_0) \in C$ showing that $p^{-1}(D) \subseteq C$. Both inclusions give the desired equality. ■

Exercise 9 Let f be the function in Example 7. If $I = (-1, 1)$, determine $f^{-1}(I)$.

Exercise 10 Let p be the function in Example 8. If

$$S = \{(x, y, z) : x^2 + y^2 + z^2 = 1\} \subseteq \mathbb{R}^3,$$

determine $p(S)$.

Injective and Surjective Functions

Definition 11

A function $f : S \rightarrow T$ is said to be *injective* or *one-to-one* if whenever $f(s_1) = f(s_2)$ then $s_1 = s_2$.

The output of an injective function uniquely corresponds to the input; that is, the only way for two outputs to be equal ($f(s_1) = f(s_2)$) is for the inputs to have also been equal ($s_1 = s_2$). This is also alluded to in the phrase *one-to-one*. If we view a function as a collection of arrows as in Figure 2, a function is injective if every element of the codomain has *at most one* arrow pointing to it.

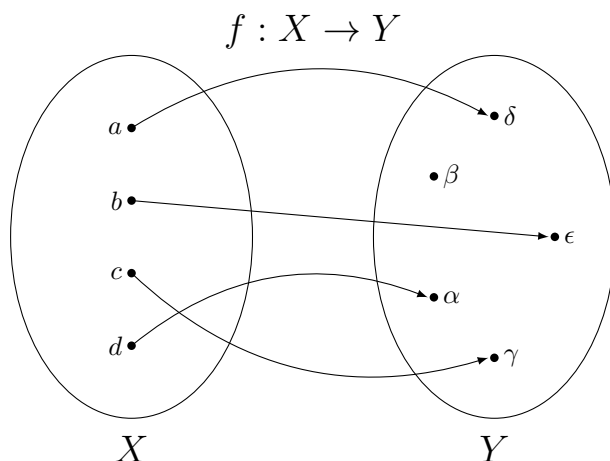


Figure 2: If $f : X \rightarrow Y$ is an injective function, each element of the codomain Y has *at most* one arrow pointing to it.

Example 12

Determine whether each of the following functions is injective.

1. $f : \{0, 1, 2, 3, 4\} \rightarrow \{0, 1, 2, 3, 4\}$ which maps n to the remainder obtained when n^3 is divided by 5.
2. $g : \mathcal{P}(\mathbb{Z}^+) \rightarrow \mathbb{Z}^+$ which maps a non-empty $S \subseteq \mathbb{Z}^+$ to its minimal element, and maps the emptyset to 1.
3. $h : (0, \infty) \rightarrow \mathbb{R}, x \mapsto x^2$.

Solution.

1. The easiest way to do this is to just check the output of f on every entry of $\{0, 1, 2, 3, 4\}$, from which we find that

$$f(0) = 0, \quad f(1) = 1, \quad f(2) = 3, \quad f(3) = 2, \quad f(4) = 4.$$

Indeed, since the output of f is unique in each case, f is injective.

2. This function is not injective, and to show it we need only provide a counterexample. Indeed, let $S_1 = \{1\}$ and $S_2 = \{1, 2\}$, for which we get $g(S_1) = 1 = g(S_2)$ despite the fact that $S_1 \neq S_2$.
3. This function is injective. Suppose that $x_1, x_2 \in (0, \infty)$ are such that $f(x_1) = f(x_2)$; that is, $x_1^2 = x_2^2$. Solving this equation gives $x_1 = \pm x_2$; however, since both numbers have to be positive, we conclude that $x_1 = x_2$, showing injectivity as required. ■

Proposition 13

If $f : B \rightarrow C$ and $g : A \rightarrow B$ are injective functions, then $h = f \circ g : A \rightarrow C$ is also injective.

Solution. Assume that $h(x) = h(y)$ for some $x, y \in A$. By definition of h we have $f(g(x)) = f(g(y))$. Since the function f is injective, the only way $f(m_1) = f(m_2)$ is if $m_1 = m_2$, so $f(g(x)) = f(g(y))$ implies that $g(x) = g(y)$. Since g is also injective, it must be the case that $x = y$. Thus we have shown that if $h(x) = h(y)$ then $x = y$, showing that h is injective as required. ■

Exercise 14 Let $f : B \rightarrow C$ and $g : A \rightarrow B$ be functions. Show that if $f \circ g$ is injective, then g is injective. Must it be true that f is injective as well?

The dual notion to an injective function is a surjective function, and this duality will be made clearer later.

Definition 15

A function $f : S \rightarrow T$ is said to be *surjective* or *onto* if for every element $t \in T$, there is an element $s \in S$ such that $f(s) = t$.

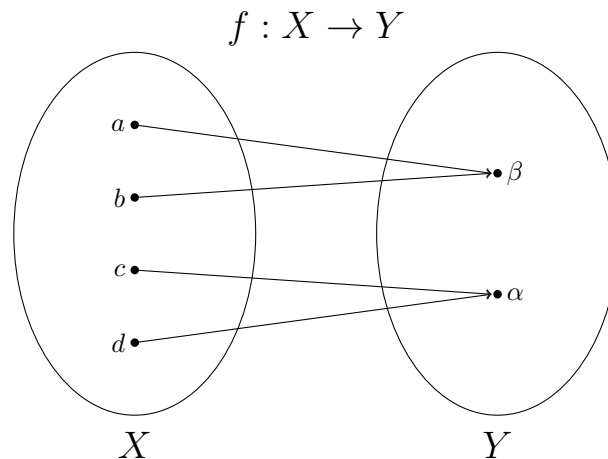


Figure 3: If $f : X \rightarrow Y$ is surjective, then every element of the codomain has *at least* one arrow pointing at it.

When thinking about surjective functions, the idea to keep in mind is that every element in T is mapped to by something in S . If we describe a function as arrows between sets, a function f is surjective if everything in T has *at least one* arrow pointing to it.

Example 16

Of the functions defined in Example 12, which are surjective?

Solution.

1. Looking at the values we've already computed in Example 12, we can see that every element of $\{0, 1, 2, 3, 4\}$ is the output of f , so f is surjective.

2. The function g is surjective. To show this, let $n \in \mathbb{Z}^+$ be arbitrary, and define $S_n = \{n\}$. Since g takes each set to its minimal element and S_n consists of a single element, we get that $g(S_n) = n$. Thus every natural number is the image of some set, showing that g is surjective.
3. This function fails to be surjective since the function h cannot hit any negative numbers. Indeed, for the sake of contradiction suppose that h is surjective. Let $x \in (0, \infty)$ be such that $h(x) = -1$. But $h(x) = x^2 > 0$ for all $x \in (0, \infty)$. No positive x can satisfy $x^2 = -1$ and hence h is not surjective. ■

Proposition 17

If $f : B \rightarrow C$ and $g : A \rightarrow B$ are surjective functions, then $f \circ g : A \rightarrow C$ is also surjective.

Solution. Let $c \in C$ be an arbitrary element, for which we need to find an $a \in A$ such that $f(g(a)) = c$. Since f is surjective, there exists $b \in B$ such that $f(b) = c$. Since g is surjective, there exists $a \in A$ such that $g(a) = b$. Now $f(g(a)) = f(b) = c$ as required. ■

Exercise 18 Determine if each function below is injective, surjective, both, or neither.

1. The function $f : \mathbb{R} \rightarrow \mathbb{R}, x \mapsto \sin(x)$.
2. The function $d : \mathbb{R} \rightarrow \mathbb{R}^2, x \mapsto (x, x)$.
3. The function $p : \mathbb{R}^2 \rightarrow \mathbb{R}, (x, y) \mapsto x$.
4. Let $\text{Poly}_{\mathbb{R}}$ be the polynomials with real coefficients, and define $\text{ev}_0 : \text{Poly}_{\mathbb{R}} \rightarrow \mathbb{R}$ as $\text{ev}_0(p) = p(0)$.

Exercise 19 Like Exercise 14, there should be a corresponding statement that goes “If $f \circ g$ is surjective, then ... ” Determine which of f or g must be surjective and prove your result.

Definition 20

A function $f : S \rightarrow T$ is bijective if it is both injective and surjective.

If $f : S \rightarrow T$ is injective, every element of T has *at most* one arrow pointing at it. If f is surjective, then every element of T has *at least* one arrow pointing at it. If f is bijective (and hence both injective and surjective), this must mean that every element of T has *exactly* one arrow pointing at it. We’ve shown that compositions of injective/surjective functions are injective/surjective, so it immediately follows that compositions of bijections are bijections.

Exercise 21 Consider the maps defined in Example 12 and Exercise 18. Which, if any, of these functions are bijective?

Exercises

1. Consider the function $f : \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = \sin(x)$. Determine $f([-2\pi, 2\pi])$ and $f^{-1}(0)$.
2. Consider the function $g : \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z}$ given by $g(x, y) = xy$. Let $U = \{(x, y) \in \mathbb{Z} \times \mathbb{Z} : x^2 + y^2 \leq 4\}$ and $\mathbb{P}$ denote the set of prime numbers. Find $g(U)$ and $g^{-1}(\mathbb{P})$.
3. Consider the function $h : \mathbb{R} \rightarrow \mathbb{R}^2$ given by $h(x) = (x^3, x^2)$. Find $h^{-1}([-1, 1]^2)$.
4. Suppose that $f : A \rightarrow B$ and $U_1, U_2 \subseteq A$. Show that $f(U_1 \cap U_2) \subseteq f(U_1) \cap f(U_2)$. Give an explicit counter-example to show that this is not an equality.
5. Suppose that $f : A \rightarrow B$ and $U_1, U_2 \subseteq A$. Show that $f(U_1 \cup U_2) = f(U_1) \cup f(U_2)$.
6. For each of the following functions, determine if they are injective, surjective, both (bijective), or neither.
 - (a) $f_1 : \mathbb{R} \rightarrow \mathbb{R}, f_1(x) = |x|$.
 - (b) $f_2 : \mathbb{Z}^+ \times \mathbb{Z}^+ \rightarrow \mathbb{Z}^+, f_2(x, y) = \min(x, y)$.
 - (c) $f_3 : \mathbb{Z}^2 \rightarrow \mathbb{Z}^2, f_3(x, y) = (xy, y^3)$.
 - (d) $f_4 : \mathbb{R} \rightarrow [0, 1], f_4(x) = \begin{cases} 1/b & x \in \mathbb{Q}, x = a/b \text{ in lowest terms with } b > 0 \\ 0 & \text{otherwise} \end{cases}$
7. Suppose that $f : A \rightarrow B$ and $g : A \rightarrow B$ are functions. Define $h : A \rightarrow B \times B$ by $h(a) = (f(a), g(a))$. Show that if f or g is injective, then h is injective. Is the converse true?
8. A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is said to be strictly increasing if $f(x) < f(y)$ whenever $x < y$. Show that a strictly increasing function is necessarily injective.
9. Let B be the set of infinite binary strings; for example, $0110100010111 \dots$. These are the strings formed from 0's and 1's which continue without end. Show that there is a bijection between B and the set $N = \{f : \mathbb{Z}^+ \rightarrow \{0, 1\}\}$.